

VICTOR VELAZQUEZ

AI Engineer | Agentic Systems | LLMs & RAG

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PROFESSIONAL SUMMARY

AI Engineer with 6+ years of experience designing and shipping production-grade agentic systems, Retrieval-Augmented Generation (RAG) pipelines, and LLM-powered applications. Proven track record building multi-agent orchestrators with LangGraph, conversational AI platforms, computer vision systems, and autonomous robotics. Combines deep mathematical foundations with full-stack engineering to deliver observable, scalable, and evaluation-driven AI solutions across conversational AI, aerospace, medical imaging, and quantitative finance domains.

CORE COMPETENCIES

Agentic AI & LLMs: LangChain, LangGraph, LangSmith, CrewAI, MCP, Multi-Agent Orchestration, Prompt Engineering, RAG, Vector Databases (Chroma, Pinecone), Embeddings, Guardrails, LLM Evaluation, Conversational AI, GPT-4, AWS Bedrock, Pydantic

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Reinforcement Learning, Deep Q-Learning, Transfer Learning, Ensemble Models, NLP, Forecasting, Classification

Computer Vision & Robotics: OpenCV, OCR, ROS, Gazebo, SLAM, Navigation, Perception, Motion Planning, LiDAR, Point-Cloud Processing, Sensor Fusion, Robotic Arms (xArm6)

Backend, Cloud & MLOps: Python, C/C++, FastAPI, Django, Flask, Docker, Linux, Git, CI/CD, AWS, Azure, REST, GraphQL, SQL, NoSQL (MongoDB), Microservices, Observability

Quantitative & Other: Modern Portfolio Theory, Algorithmic Trading, PineScript, Matlab/Simulink, Tableau, PowerBI, SolidWorks, Scrum/Agile

PROFESSIONAL EXPERIENCE

AI/ML Engineer | A2C | Remote

Jan 2026 - Present

- Architected production multi-agent orchestration systems using LangGraph and LangSmith, reducing manual decision-routing time by approximately 40% across internal business workflows.
- Built and deployed RAG pipelines integrating GPT-4 with Chroma and Pinecone vector stores, serving 10K+ document queries per week with sub-2s p95 latency.
- Implemented LLM evaluation harness with automated regression tests, CI gates, and observability dashboards covering 50+ prompt variants and retrieval scenarios.
- Designed guardrails, policy layers, and groundedness checks for agentic workflows, cutting hallucination rate by approximately 35% on internal benchmarks.

AI Engineer | El Paso Labs | Remote

Jun 2025 - May 2026

- Delivered AtlasOps, a multi-agent decision and retrieval orchestrator built on LangGraph with persistent memory, guardrails, and full observability; adopted across 3 internal business units.
- Engineered RAG-based document Q&A platform with FastAPI, LangChain, and ChromaDB, reducing time-to-answer for internal policy queries from hours to under 30 seconds.
- Built agentic process automation pipelines integrating MCP services and tool-using agents, eliminating ~60% of repetitive operational tasks for stakeholders.
- Authored evaluation suites for prompt regression and retrieval quality (precision@k, recall@k, faithfulness, groundedness) integrated into CI/CD.

Software Engineer | Dynamic Optics | Padova, Italy

2024 - 2025

- Developed an Adaptive Optics control library for the European Space Agency (ESA) enabling satellite-to-telescope laser communication under high atmospheric turbulence.
- Implemented C++/Python control algorithms, hardware abstraction layers, and operations tooling validated through integration testing on flight-grade equipment.

Deep Learning Researcher | University of Padova | Padova, Italy

2024

- Designed PyTorch ensemble segmentation model for medical imaging assistance, combining 4 SOTA architectures with transfer learning across heterogeneous datasets.
- Improved generalization by approximately 18% over single-model baselines through targeted data augmentation and weighted ensembling; published as M.Sc. thesis.

Robotics Software Engineer | Subdron | Bregenz, Austria

2023

- Developed 3D point-cloud processing pipeline and ROS-based noise-filtering modules for underwater autonomous inspection vehicles, enabling robust SLAM in turbid water.
- Conducted on-mission robot operation, sensor calibration, and field maintenance during commercial deployments.

Machine Learning Engineer | Natma | Mexico City, Mexico

2020 - 2022

- Built end-to-end ML pipelines for forecasting, classification, and computer vision using PyTorch, TensorFlow, and OpenCV; deployed via FastAPI, Docker, and AWS CI/CD.
- Designed dataset exploration and ETL workflows on AWS, reducing data preparation time by approximately 50% across recurring projects.

Quantitative Developer & Portfolio Analyst | LatamLead | Remote

2019 - Present

- Implemented automated investing systems applying Modern Portfolio Theory, factor analysis, and quantitative risk modeling; coded backtested strategies in PineScript and Python.

FEATURED PROJECTS

AtlasOps - Multi-Agent Decision & Retrieval Orchestrator

Stack: LangGraph, LangSmith, Python, FastAPI, Docker

Production-oriented multi-agent system that plans, retrieves, reasons, and executes on real business questions with guardrails, persistent memory, CI tests, and full observability.

Intelligent Document Q&A - RAG Platform

Stack: LangChain, GPT-4, ChromaDB, FastAPI

Retrieval-Augmented Generation pipeline with hybrid search, source citation, and prompt-evaluation harness for internal policy and document Q&A.

Adaptive Optics Control Library - ESA Project

Stack: Python, C++, Modbus, Hardware Integration

Software library enabling satellite-to-telescope laser communication under high atmospheric turbulence; deployed in partnership with the European Space Agency.

Autonomous Automotive Rear Spoiler - Deep Q-Learning

Stack: Keras, Matlab, SolidWorks, Embedded Sensors

Intelligent control of vehicle rear spoiler for lateral acceleration mitigation. Awarded Finalist - Best Paper, Japan Mechatronics Conference.

EDUCATION

M.Sc. Computer Engineering | University of Padova, Italy | 2022 - 2024

B.Sc. Mechanical Electrical Engineering | Tecnologico de Monterrey, Mexico | 2016 - 2020

International Exchange Program | Universidad Politecnica de Catalunya, Spain | 2019

Data Science Nano Degree | Acamica | 2021

CERTIFICATIONS

- LangChain AI Developer Masterclass - Udemy (2025)
- SQL for Data Science - University of California (2023)
- Self-Driving Cars Specialization - University of Toronto (2023)
- Algorithmic Trading Python + AWS - Udemy (2022)
- AWS Fundamentals - Amazon Web Services (2021)
- Natural Language Processing Specialization - DeepLearning.AI (2020)
- Deep Learning Specialization - DeepLearning.AI (2020)
- Deep Reinforcement Learning - SuperDataScience (2020)

AWARDS & RECOGNITION

- Finalist, Best Paper - Japan Mechatronics Conference (B.Sc. thesis on Reinforcement-Learning-based active aerodynamics).

LANGUAGES

Spanish (Native) | English (Professional) | Italian (Working)